THQ #8

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Section 6.2

Question 8

(a)

Let $x_i \in \{0,1\}$ for $i = 1, \dots, n$, where $x_i = 1$ if item i is taken and $x_i = 0$ otherwise.

$$\begin{aligned} \max \quad & \sum_{i=1}^{n} c_i x_i \\ \text{s.t.} \quad & \sum_{i=1}^{n} a_i x_i \le V \\ & \sum_{i=1}^{n} w_i x_i \le W \\ & x_i \in \{0,1\}, \quad i = 1, \dots, n. \end{aligned}$$

(b)

Same decision variables, objective, and capacity constraints as in (a), plus:

$$\begin{aligned} & x_1 \le x_4, \quad x_2 \le x_4, \quad x_3 \le x_4 && \text{(cans need can opener)} \\ & x_5 + x_6 = 1 && \text{(exactly one snack)} \\ & x_2 \le x_7, \quad x_3 \le x_7, \quad x_6 \le x_7 && \text{(items using cooking pot)} \\ & x_i \in \{0,1\}, \quad i = 1, \dots, n. \end{aligned}$$

Question 9

Let - x_s = number of 10-week state jobs, - x_c = number of 6-week county jobs, - x_d = number of 3-week developer jobs, - x_p = number of weeks of parking-lot work, - $y \in \{0,1\}$, where $y = 1$ if any state or county work is done (so the \$7500 contribution is paid).

$$\begin{aligned} \max \quad & 32000\,x_s + 17400\,x_c + 8250\,x_d + 2550\,x_p - 7500\,y \\ \text{s.t.} \quad & 10x_s + 6x_c + 3x_d + x_p \le 28 \\ & x_s + x_c \le 4y \\ & x_s, x_c, x_d, x_p \ge 0 \text{ and integer} \\ & y \in \{0,1\}. \end{aligned}$$

(Here 4 is a valid big-M: at most four state/county jobs fit in 28 weeks, e.g. four 6-week county jobs.)

Question 15

Let $y_1, y_2 \in \{0,1\}$, where $y_1 = 1$ if the first constraint set is enforced and $y_2 = 1$ if the second is enforced.

Estimating the M_i (slide 24 style). From the $\le$ constraints, one variable at a time:

- $x_1 + x_3 \le 100 \to x_1 \le 100$, $x_3 \le 100$
- $2x_1 - 4x_2 + 5x_3 \le 250 \to x_1 \le 125$, $x_3 \le 50$
- $5x_1 + 4x_2 + 2x_3 \le 300 \to x_1 \le 60$, $x_2 \le 75$, $x_3 \le 150$

Section 6.2:

Formulate integer (or mixed integer) programming models for the following. (Do not attempt to solve the problems.)

8. (a) (*The Knapsack Problem*) A backpacker's knapsack has a volume of V in.^{3} and can hold up to W lb of gear. The backpacker has a choice of n items to carry in it, with the ith item requiring a_i in.^{3} of space, weighing w_i lb, and providing c_i units of value for the trip. What items should be taken in the knapsack?

(b) Refine part (a) to include the following considerations: Item 1, a can of tuna fish, Item 2, a can of corn, and Item 3, a can of stew, have no value unless Item 4, the can opener, is taken; and only one snack, either Item 5, potato chips (light but bulky), or Item 6, unpopped popcorn (small but heavy), is to go. Of course Items 2, 3, and 6 all use Item 7, the cooking pot.

9. A road construction firm seeks to assign its force optimally over the next 28-week period. They can be assigned to any combination of the following:

- For any number of 10-week periods, working for the state, and earning a profit of \$3200/week.
- For any number of 6-week periods, working for the county, and earning a profit of \$2900/week.
- For any number of 3-week periods, working for a private land developer, and earning \$2750/week.
- For any number of weeks, working on parking lot construction, and earning \$2550/week.
- However, if the firm does any work at all for either the state or county (or both), it is expected to contribute \$7500 to the campaign fund of a certain anonymous political figure.

15. Maximize $3x_1 + 5x_2 + 7x_3$ subject to

$5x_1 + 4x_2 + 2x_3 \leq 300, x_1, x_2, x_3 \geq 0$, and

$$\left\{ \begin{array}{rl} x_1 + x_3 \leq & 100 \\ x_1 - x_2 \geq & 0 \end{array} \right\} \quad \text{or} \quad \left\{ \begin{array}{rl} 2x_1 - 4x_2 + 5x_3 \leq & 250 \\ x_2 - 2x_3 \geq & 50 \end{array} \right\}$$

Figure 1: Section 6.2 problems

Largest caps: $x_1 \le 125$, $x_2 \le 75$, $x_3 \le 150$ (and $x_i \ge 0$). With each set off ($y = 0$):

- M_a: $x_1 + x_3 - 100 \le 125 + 150 - 100 = 175 \to M_a = 175$
- M_b: $x_1 - x_2 \ge -M_b \to M_b \ge 75 - 0 = 75 \to M_b = 75$
- M_c: $2x_1 - 4x_2 + 5x_3 - 250 \le 2(125) + 5(150) - 250 = 750 \to M_c = 750$
- M_d: $x_2 - 2x_3 \ge 50 - M_d \to M_d \ge 50 - (0 - 2 \cdot 150) = 350 \to M_d = 350$

$$
\begin{aligned}
\max \quad & 3x_1 + 5x_2 + 7x_3 \\
\text{s.t.} \quad & 5x_1 + 4x_2 + 2x_3 \le 300 \\
& x_1 + x_3 \le 100 + 175(1 - y_1) \\
& x_1 - x_2 \ge -75(1 - y_1) \\
& 2x_1 - 4x_2 + 5x_3 \le 250 + 750(1 - y_2) \\
& x_2 - 2x_3 \ge 50 - 350(1 - y_2) \\
& y_1 + y_2 = 1 \\
& x_1, x_2, x_3 \ge 0 \\
& y_1, y_2 \in \{0, 1\}.
\end{aligned}
$$

When $y_1 = 1$ (so $y_2 = 0$), the first pair is active and the second is relaxed; when $y_2 = 1$, the reverse holds.

Section 6.3

Question (a)

Introduce slack variables $x_3, x_4 \ge 0$:

$$
\begin{aligned}
\min \quad & x_1 - x_2 \\
\text{s.t.} \quad & 3x_1 + 4x_2 + x_3 = 6 \\
& x_1 - x_2 + x_4 = 1 \\
& x_1, x_2, x_3, x_4 \ge 0 \text{ and } x_1, x_2 \text{ integral.}
\end{aligned}
$$

Initial tableau (same layout as Table 6.2; x_3, x_4 basic):

	x_1	x_2	x_3	x_4	
x_3	3	4	1	0	6
x_4	1	−1	0	1	1
	1	−1	0	0	0

Bottom-row x_2 entry is negative → enter x_2. Only positive pivot candidate in that column is 4 → exit x_3. Divide the x_3 row by 4, then clear the x_2 column in the x_4 and z rows:

	x_1	x_2	x_3	x_4	
x_2	$\frac{3}{4}$	1	$\frac{1}{4}$	0	$\frac{3}{2}$
x_4	$\frac{7}{4}$	0	$\frac{1}{4}$	1	$\frac{5}{2}$
	$\frac{7}{4}$	0	$\frac{1}{4}$	0	$\frac{3}{2}$

(x_2 row = $x_3/4$; new x_4 = old x_4+ new x_2; new z = old z+ new x_2.)

LP solution: $(x_1, x_2) = (0, \frac{3}{2})$, with min = $-\frac{3}{2}$ (tableau RHS $\frac{3}{2}$ is the max-form value of $-x_1 + x_2$). Not integer → need a cut next.

Section 6.3:

Solve the following using the Cutting Plane Algorithm.

(a) Minimize $x_1 - x_2$

subject to

$$3x_1 + 4x_2 \leq 6$$
$$x_1 - x_2 \leq 1$$
$$x_1, x_2 \geq 0 \text{ and integral}$$

(c) Maximize $2x_1 - 4x_2 + x_3$

subject to

$$x_1 - x_2 \leq 12$$
$$2x_2 + 3x_3 \leq 28$$
$$x_1, x_2, x_3 \geq 0 \text{ and integral}$$

Figure 2: Section 6.3 problems

Gomory cut from the x_2 row (slide style: integers on one side, fractional on the other):

Step 1. Source equation:

$$\frac{3}{4}x_1 + x_2 + \frac{1}{4}x_3 = \frac{3}{2}.$$

Step 2. Split each entry as $[\cdot] + f$ (floor + fractional part in $[0, 1)$):

$$\frac{3}{4} = 0 + \frac{3}{4}, \quad 1 = 1 + 0, \quad \frac{1}{4} = 0 + \frac{1}{4}, \quad \frac{3}{2} = 1 + \frac{1}{2}.$$

So

$$x_2 + \underbrace{\frac{3}{4}x_1 + \frac{1}{4}x_3}_{\text{fractional coeffs}} = 1 + \frac{1}{2}.$$

Step 3. Move integers to the right, fractionals to the left (RHS fractional part stays with the fractional left-hand side):

$$\frac{3}{4}x_1 + \frac{1}{4}x_3 - \frac{1}{2} = 1 - x_2.$$

Right-hand side is integer when x_2 is integer; left-hand side carries the fractional pieces.

Step 4. Name the left-hand side x_5 (new slack / cut variable):

$$x_5 = \frac{3}{4}x_1 + \frac{1}{4}x_3 - \frac{1}{2}.$$

Since $x_1, x_3 \geq 0$, we have $x_5 \geq -\frac{1}{2}$. With x_5 integer, $x_5 \geq 0$.

Step 5. Rewrite with $+1$ coefficient on x_5 (tableau-ready cut row):

$$-\frac{3}{4}x_1 - \frac{1}{4}x_3 + x_5 = -\frac{1}{2}.$$

Append to the current tableau (zeros under basic x_2, x_4 in the new row; z unchanged for now):

	x_1	x_2	x_3	x_4	x_5	
x_2	$\frac{3}{4}$	1	$\frac{1}{4}$	0	0	$\frac{3}{2}$
x_4	$\frac{7}{4}$	0	$\frac{1}{4}$	1	0	$\frac{5}{2}$
x_5	$-\frac{3}{4}$	0	$-\frac{1}{4}$	0	1	$-\frac{1}{2}$
	$\frac{7}{4}$	0	$\frac{1}{4}$	0	0	$\frac{3}{2}$

Dual simplex: all bottom-row $c_j \geq 0$, but b for x_5 is negative.

- **Exit:** most negative RHS $\rightarrow x_5$ $(-\frac{1}{2})$.
- **Enter:** among $a_{rj} < 0$ in that row, smallest $|c_j/a_{rj}|$:
  - x_1: $|(\frac{7}{4})/(-\frac{3}{4})| = \frac{7}{3}$
  - x_3: $|(\frac{1}{4})/(-\frac{1}{4})| = 1 \rightarrow$ enter x_3, pivot on $-\frac{1}{4}$.

Divide the x_5 row by $-\frac{1}{4}$, then clear the x_3 column:

	x_1	x_2	x_3	x_4	x_5	
x_2	0	1	0	0	1	1
x_4	1	0	0	1	1	2
x_3	3	0	1	0	-4	2
	1	0	0	0	1	1

All RHS ≥ 0 and bottom row $\geq 0 \rightarrow$ optimal. Integer solution:

$$(x_1, x_2) = (0, 1), \qquad \min\ x_1 - x_2 = -1.$$

Question (c)

Introduce slack variables $x_4, x_5 \geq 0$:

$$\begin{aligned} \max \quad & 2x_1 - 4x_2 + x_3 \\ \text{s.t.} \quad & x_1 - x_2 + x_4 = 12 \\ & 2x_2 + 3x_3 + x_5 = 28 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \text{ and } x_1, x_2, x_3 \text{ integral.} \end{aligned}$$

Initial tableau (bottom row $=-$ max coefficients, same convention as (a)):

	x_1	x_2	x_3	x_4	x_5	
x_4	1	-1	0	1	0	12
x_5	0	2	3	0	1	28
	-2	4	-1	0	0	0

Enter x_1 (most negative bottom entry). Pivot on 1 in the x_4 row:

	x_1	x_2	x_3	x_4	x_5	
x_1	1	-1	0	1	0	12
x_5	0	2	3	0	1	28
	0	2	-1	2	0	24

Enter x_3. Pivot on 3 in the x_5 row:

	x_1	x_2	x_3	x_4	x_5	
x_1	1	-1	0	1	0	12
x_3	0	$\frac{2}{3}$	1	0	$\frac{1}{3}$	$\frac{28}{3}$
	0	$\frac{8}{3}$	0	2	$\frac{1}{3}$	$\frac{100}{3}$

LP optimal at $(x_1, x_2, x_3) = (12,\ 0,\ \frac{28}{3})$, $z = \frac{100}{3}$. Not integer $\rightarrow$ Gomory cut from the x_3 row.

Gomory cut. Source row:

$$\frac{2}{3}x_2 + x_3 + \frac{1}{3}x_5 = \frac{28}{3} = 9 + \frac{1}{3}.$$

Fractional left, integers right:

$$\frac{2}{3}x_2 + \frac{1}{3}x_5 - \frac{1}{3} = 9 - x_3.$$

Let x_6 be the cut slack:

$$-\frac{2}{3}x_2 - \frac{1}{3}x_5 + x_6 = -\frac{1}{3}.$$

	x_1	x_2	x_3	x_4	x_5	x_6	
x_1	1	-1	0	1	0	0	12
x_3	0	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	$\frac{28}{3}$
x_6	0	$-\frac{2}{3}$	0	0	$-\frac{1}{3}$	1	$-\frac{1}{3}$
	0	$\frac{8}{3}$	0	2	$\frac{1}{3}$	0	$\frac{100}{3}$

Dual simplex: exit x_6. Ratios for $a_{rj} < 0$: - x_2: $|(\frac{8}{3})/(-\frac{2}{3})| = 4$ - x_5: $|(\frac{1}{3})/(-\frac{1}{3})| = 1 \to$ enter x_5, pivot on $-\frac{1}{3}$:

	x_1	x_2	x_3	x_4	x_5	x_6	
x_1	1	-1	0	1	0	0	12
x_3	0	0	1	0	0	1	9
x_5	0	2	0	0	1	-3	1
	0	2	0	2	0	1	33

All RHS ≥ 0 and bottom row $\geq 0 \to$ optimal integer solution:

$$(x_1, x_2, x_3) = (12,\ 0,\ 9), \qquad \max\ 2x_1 - 4x_2 + x_3 = 33.$$